RevConnect Project

Full-Stack Professional Networking Platform

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Introduction & Objective

RevConnect is a digital ecosystem designed to build connections, share content, and support professional growth.

Key Objectives

- Develop a scalable full-stack web application.
- Implement secure JWT authentication.
- Design a normalized relational database architecture.

System Architecture

- **Presentation Layer**
  Angular (Component-based UI), TypeScript, HTML/CSS

- **Application Layer**
  Spring Boot, REST APIs, Spring Security (JWT)

- **Data Layer**
  MySQL Relational Database, Spring Data JPA / Hibernate

Methodology

1. **Requirement & DB Design**
   Normalized schema creation

2. **Backend Build**
   Spring Boot APIs & Security

3. **Frontend UI**
   Angular component integration

Database Design Overview

- **18+ Relational Tables:** Highly normalized schema designed for efficiency and scalability.
- **Core Entities:** Users, Profiles, Posts, Comments, and Likes.
- **Networking Logic:** Connections, Follows, and Reposts tables to map interactions.
- **Business Logic:** Dedicated tables for Business Profiles, Operating Hours, Products, and External Links.
- **Relationships:** Robust implementation of one-to-many and many-to-many associations.

Core Modules: Personal Users

- **Auth & Profile**
  Registration, secure login, and profile editing.
- **Post Management**
  Create, edit, delete posts, and use hashtags.
- **Social Interaction**
  Like, comment, share, and follow accounts.
- **Notifications**
  In-app alerts for interactions and connections.

Creator & Business Features

- **Enhanced Profiles:** Add business hours, location, and links.
- **Content Strategy:** Create promotional posts and tag products.
- **Advanced Tools:** Schedule posts and pin important updates.
- **Analytics:** View likes, comments, shares, and reach.

Findings & Observations

- **Security:** JWT significantly improves session security.
- **Communication:** REST architecture simplifies API integration.
- **Data Structure:** Proper relational mapping is essential for feeds.
- **Maintainability:** Modular design improves long-term scalability.

Challenges & Key Learning

@ Challenges Faced

- Integration: Connectiong Angular dynamically with Spring Boot RESR APIs securely.
- Data Optimization: Handling complex relational JOINs for personalized feed generation efficiently.

Key Learning

- Gained a comprehensive understanding of the entire Full-Stack development Lifecycle.
- Mastered Database relationships modelling for scaling social applications.
- Improved overall debugging & problem-solving skills across isolated application layers.

Conclusion

”RevConnect successfully bridges the gap between personal social networking and business engagement, providing a unified, secure, and scalable digital platform for professional communication.”

Thank You!